

Walchand College of Engineering, Sangli
Computer Science & Engineering
Third Year
Course: Computer Algorithm Lab

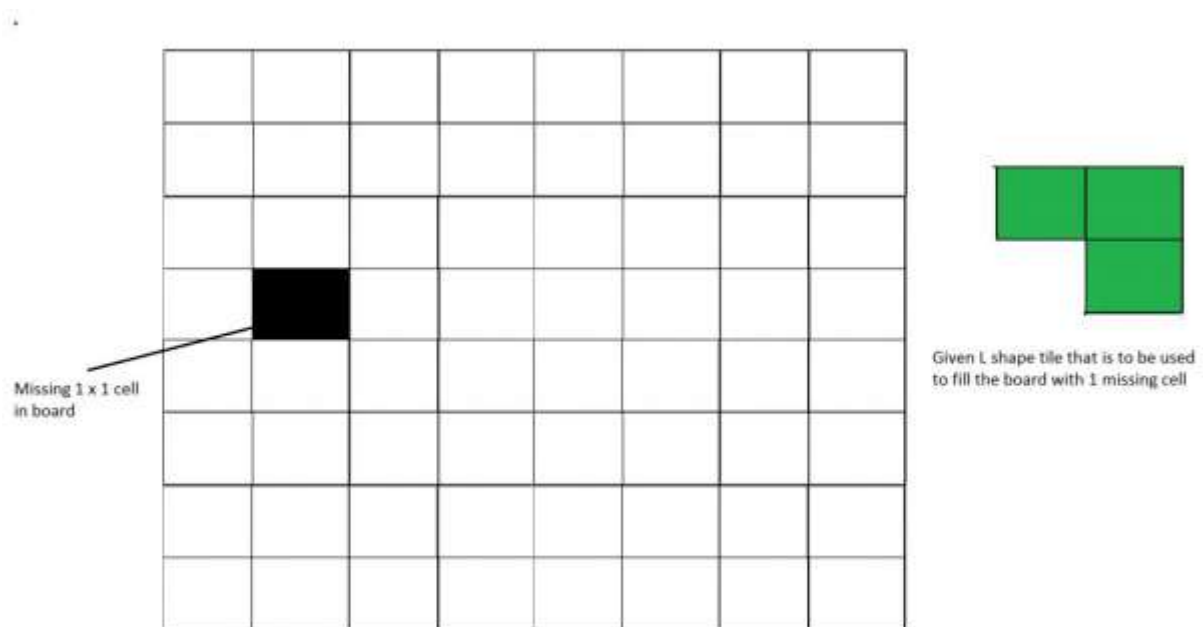
Week 4 Assignment

Part: 1

Divide and conquer strategy

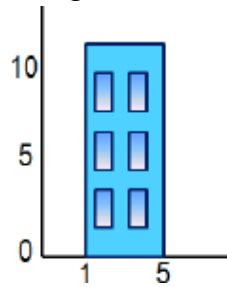
Q1) Implement algorithm to Find the maximum element in an array which is first increasing and then decreasing, with Time Complexity $O(\log n)$.

Q2) Implement algorithm for Tiling problem: Given an $n \times n$ board where n is of form 2^k where $k \geq 1$ (Basically n is a power of 2 with minimum value as 2). The board has one missing cell (of size 1×1). Fill the board using L shaped tiles. An L shaped tile is a 2×2 square with one cell of size 1×1 missing



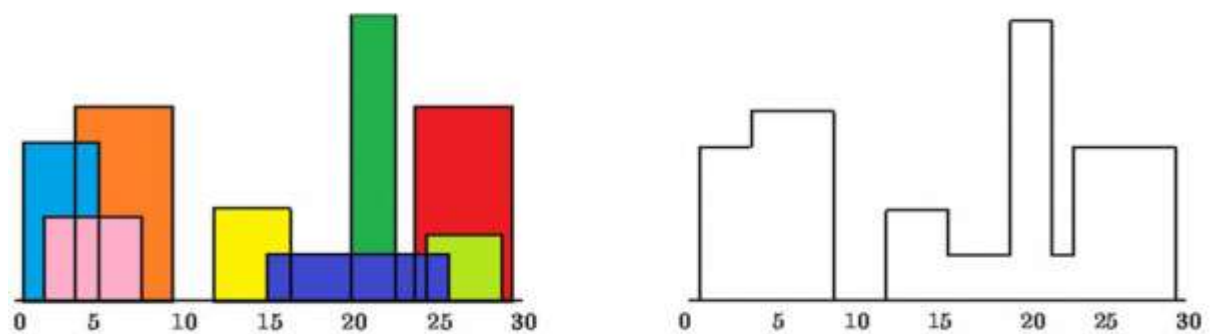
Q3) Implement algorithm for The Skyline Problem: Given n rectangular buildings in a 2-dimensional city, computes the skyline of these buildings, eliminating hidden lines. The main task is to view buildings from a side and remove all sections that are not visible.

All buildings share common bottom and every building is represented by triplet (left, ht, right) 'left': is x coordinated of left side (or wall). 'right': is x coordinate of right side 'ht': is height of building. For example, the building on right side is represented as (1, 11, 5)



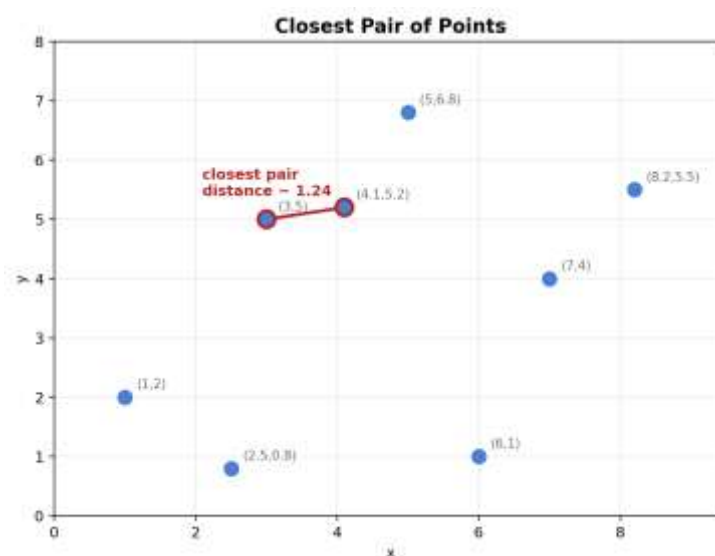
A **skyline** is a collection of rectangular strips. A rectangular **strip** is represented as a pair (left, ht) where left is x coordinate of left side of strip and ht is height of strip.

With Time Complexity $O(n \log n)$



Q4) Implement an algorithm using Divide and Conquer to find the pair of points with the smallest Euclidean distance among n points given in a 2-dimensional plane.

Given the (x, y) coordinates of n points, find the two points that are closest to each other and report the minimum distance between them. With Time Complexity $O(n \log n)$



Q5) Implement Strassen's algorithm using Divide and Conquer to multiply two $n \times n$ matrices A and B, where n is a power of 2, using fewer recursive sub-multiplications than the straightforward block-multiplication approach. With Time Complexity $O(n^{\log_2(7)})$, i.e. approximately $O(n^{2.81})$

Note: You may refer to similar problems on platforms such as LeetCode or CodeChef to understand the problem statement, but your implementation, choice of pivot/split strategy, and analysis must be your own. Treat the sample inputs above only as illustrations of the expected input and output format — your program should work correctly for any valid input of these types, not just the example shown. At least 4 test case should be available in the output document.